Analysis of Review Length and Sentiment

JIAXING HE

Introduction

Objective:

Analyze relationship between review length and sentiment

Datasets:

Amazon, IMDb, Steam

Methods:

Review length
Sentiment label (0: negative; 1: positive)
Correlation analysis (point-biserial correlation)

Data Source

#	Name	Source URL	Type	Fields	Format	Data Size
1	Amazon Fine Food Reviews	https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews	File	Review, Scoring	csv	500,000
2	IMDB Movie Reviews	https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews	File	Review, Sentiment	csv	50,000
3	Steam Reviews (ID: 2868840)	https://store.steampowered.com/appreviews/2868840	API	Review, Label	JSON	600
4	Steam Reviews (ID: 570)	https://store.steampowered.com/appreviews/570	API	Review, Label	JSON	600
5	Steam Reviews (ID: 1172470)	https://store.steampowered.com/appreviews/1172470	API	Review, Label	JSON	600

Results(Amazon)

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Amazon Analysis
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Label counts:
label
0    400
1    400
Name: count, dtype: int64

Review length statistics by label:
       count     mean        std   min   25%   50%     75%     max
label
0      400.0  88.5075  93.215760  12.0  39.0  64.5  103.00  1027.0
1      400.0  80.2700  79.605116   6.0  32.0  57.0  100.25   701.0

Point-biserial correlation between review_length and label:
Correlation: -0.0475
P-value: 0.1793
Interpretation: Longer reviews tend to be more negative.
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Correlation: close to 0

p-value: > 0.05

Not statistically significant

Results(IMDB)

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IMDb Analysis
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Label counts:
label
0    300
1    300
Name: count, dtype: int64

Review length statistics by label:
       count        mean         std   min     25%    50%     75%    max
label
0      300.0  213.576667  146.644307  25.0  123.75  168.5  254.00  977.0
1      300.0  209.406667  158.173589  40.0  119.75  157.0  247.25  980.0

Point-biserial correlation between review_length and label:
Correlation: -0.0137
P-value: 0.7378
Interpretation: Longer reviews tend to be more negative.
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Correlation: close to 0

P-value: >0.05

Not statistically significant

Results(Steam)

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Steam Analysis
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Label counts:
label
0    294
1    275
Name: count, dtype: int64

Review length statistics by label:
       count       mean        std  min   25%   50%    75%    max
label
0      294.0  76.166667  102.56437  1.0  17.0  42.0  93.75  728.0
1      275.0  35.025455   49.86159  1.0   8.0  19.0  40.00  381.0

Point-biserial correlation between review_length and label:
Correlation: -0.2451
P-value: 0.0000
Interpretation: Longer reviews tend to be more negative.
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Correlation: -0.2451

P-value: < 0.05

Longer reviews tend to be more negative.

Results(Steam)

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Steam Analysis
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Label counts:
label
0    178
1    127
Name: count, dtype: int64

Review length statistics by label:
       count       mean        std  min   25%   50%   75%    max
label
0      178.0  32.938202  48.600081  1.0  6.25  14.0  31.0  340.0
1      127.0  18.007874  31.259156  1.0  4.00   8.0  18.0  198.0

Point-biserial correlation between review_length and label:
Correlation: -0.1721
P-value: 0.0026
Interpretation: Longer reviews tend to be more negative.
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Steam Analysis
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Label counts:
label
0    226
1    147
Name: count, dtype: int64

Review length statistics by label:
       count       mean        std  min  25%   50%   75%    max
label
0      226.0  38.530973  69.723335  1.0  6.0  14.0  33.0  554.0
1      147.0  21.938776  32.189130  1.0  5.0  11.0  21.5  194.0

Point-biserial correlation between review_length and label:
Correlation: -0.1390
P-value: 0.0072
Interpretation: Longer reviews tend to be more negative.
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User behaviors may be different across platforms. For steam particularly, longer reviews tended to be more negative.

Challenges

Scraping Steam Data:

- Languages
- Data

Thank You!